

# Amazon E-Commerce Sales Performance

Data Analytics & Business Intelligence Report | Author: Data Analyst Portfolio Project  
Tech Stack: Python (Pandas, NumPy, Matplotlib, Seaborn)

## EXECUTIVE SUMMARY & BUSINESS OVERVIEW

This report delivers an end-to-end exploratory data analysis of **185,000+ sales transactions** across the United States. Utilizing Python, key financial metrics, customer buying temporal patterns, regional demand distribution, and product price-volume relationships were modeled to drive high-impact strategic business decisions.

### Core Business Highlights:

- **Total Revenue Generated:** \$34.47 Million (\$34,465,000 / 344.65 Lakhs) across 208,812 units sold.
- **Top Revenue Driver:** Macbook Pro Laptop (\$8.03M / 23.3% total revenue).
- **Top Volume Driver:** AAA Batteries 4-Pack (30,986 units) & AA Batteries 4-Pack (27,615 units).
- **Dominant Category:** Laptops (\$12.16M / 35.3%) & Smartphones (\$8.94M / 26.0%).
- **Geographic Focus:** San Francisco, Los Angeles, and New York City account for **53.3% (\$18.37M)** of total sales.
- **Peak Shopping Window:** Q4 Holiday Season (\$11.54M / 33.5%), with daily peaks at 12:00 PM and 7:00 PM.

## 1. High-Level Key Performance Indicators (KPIs)

| Metric                 | Value                       | Share / Benchmark        |
|------------------------|-----------------------------|--------------------------|
| Total Revenue          | \$34,465,000 (\$34.47M)     | 100.0%                   |
| Total Quantity Sold    | 208,812 units               | 100.0%                   |
| Total Orders Processed | 185,686 orders              | Avg ~1.12 units/order    |
| Top Revenue Category   | Laptop (\$12,160,000)       | 35.28% of total revenue  |
| Top Volume Category    | Battery (58,601 units)      | 28.06% of total volume   |
| Top Revenue Product    | Macbook Pro (\$8,032,000)   | 23.30% of total revenue  |
| Top Revenue City       | San Francisco (\$8,255,000) | 23.95% of total revenue  |
| Peak Sales Month       | December (\$4,608,000)      | 13.37% of annual revenue |

## 2. Detailed Product-Level Performance Matrix

The analysis highlights a sharp revenue concentration in premium computing and mobile hardware alongside high-volume sales of low-cost utility accessories.

| Product Name             | Category   | Price (\$) | Qty Sold | Total Sales (\$) | Share (%) |
|--------------------------|------------|------------|----------|------------------|-----------|
| Macbook Pro Laptop       | Laptop     | \$1,700    | 4,725    | \$8,032,000      | 23.30%    |
| iPhone                   | Phone      | \$700      | 6,847    | \$4,793,000      | 13.91%    |
| ThinkPad Laptop          | Laptop     | \$1,000    | 4,128    | \$4,128,000      | 11.98%    |
| Google Phone             | Phone      | \$600      | 5,529    | \$3,317,000      | 9.62%     |
| 27in 4K Gaming Monitor   | Monitor    | \$390      | 6,239    | \$2,433,000      | 7.06%     |
| 34in Ultrawide Monitor   | Monitor    | \$380      | 6,192    | \$2,353,000      | 6.83%     |
| Apple AirPods Headphones | Headphones | \$150      | 15,637   | \$2,346,000      | 6.81%     |
| Flatscreen TV            | TV         | \$300      | 4,813    | \$1,444,000      | 4.19%     |

|                            |                |       |        |             |       |
|----------------------------|----------------|-------|--------|-------------|-------|
| Bose SoundSport Headphones | Headphones     | \$100 | 13,430 | \$1,343,000 | 3.90% |
| 27in FHD Monitor           | Monitor        | \$150 | 7,541  | \$1,131,000 | 3.28% |
| Vareebadd Phone            | Phone          | \$400 | 2,068  | \$827,000   | 2.40% |
| 20in Monitor               | Monitor        | \$110 | 4,126  | \$454,000   | 1.32% |
| LG Washing Machine         | Home Appliance | \$600 | 666    | \$400,000   | 1.16% |
| LG Dryer                   | Home Appliance | \$600 | 646    | \$388,000   | 1.13% |
| Lightning Charging Cable   | Charging Cable | \$15  | 23,169 | \$346,000   | 1.00% |
| USB-C Charging Cable       | Charging Cable | \$12  | 23,931 | \$286,000   | 0.83% |
| Wired Headphones           | Headphones     | \$12  | 20,524 | \$246,000   | 0.71% |
| AA Batteries (4-pack)      | Battery        | \$4   | 27,615 | \$106,000   | 0.31% |
| AAA Batteries (4-pack)     | Battery        | \$3   | 30,986 | \$93,000    | 0.27% |

### 3. Category & Pricing Tier Analysis

| Category       | Avg Price (\$) | Quantity Sold | Total Revenue (\$) | Revenue Share (%) |
|----------------|----------------|---------------|--------------------|-------------------|
| Laptop         | \$1,373.53     | 8,853         | \$12,160,000       | 35.28%            |
| Phone          | \$618.78       | 14,444        | \$8,938,000        | 25.93%            |
| Monitor        | \$264.60       | 24,098        | \$6,371,000        | 18.49%            |
| Headphones     | \$81.48        | 49,591        | \$3,934,000        | 11.41%            |
| TV             | \$300.00       | 4,813         | \$1,444,000        | 4.19%             |
| Home Appliance | \$600.00       | 1,312         | \$787,000          | 2.28%             |
| Charging Cable | \$13.44        | 47,100        | \$632,000          | 1.83%             |
| Battery        | \$3.41         | 58,601        | \$199,000          | 0.58%             |

#### Price Tier Insights:

- **Premium Tier (> \$500):** Generates **\$21.06M (61.1%)** of total revenue, driven by Laptops & Flagship Phones.
- **Mid Range Tier (\$200 - \$500):** Generates **\$7.06M (20.5%)**, supported by Gaming Monitors & TVs.
- **Affordable Tier (\$50 - \$200):** Generates **\$5.27M (15.3%)**, led by Wireless Headphones & FHD Monitors.
- **Budget Tier (< \$50):** Generates **\$1.08M (3.1%)**, despite driving over 52% of total transaction volume.

### 4. Temporal & Seasonal Performance Analysis

| Quarter        | Total Sales (\$) | Share (%) | Key Driver                                        |
|----------------|------------------|-----------|---------------------------------------------------|
| Q1 (Jan - Mar) | \$6,826,500      | 19.81%    | Post-holiday re-baseline; March growth rebound    |
| Q2 (Apr - Jun) | \$9,116,100      | 26.45%    | Spring promotional push & April demand spike      |
| Q3 (Jul - Sep) | \$6,982,000      | 20.26%    | Summer slowdown; September seasonal low (\$2.09M) |
| Q4 (Oct - Dec) | \$11,540,900     | 33.49%    | Holiday shopping peak (Dec \$4.61M & Oct \$3.74M) |

#### Hourly & Daily Buying Patterns:

- **Dual Hourly Peaks:** Purchasing spikes sharply during **11:00 AM - 1:00 PM** (12 PM peak: \$2.31M sales, 12,080 orders) and **6:00 PM - 8:00 PM** (7 PM peak: \$2.41M sales, 12,380 orders).
- **Time Slot Dominance:** Afternoon (\$10.39M) and Evening (\$9.04M) account for **56.4%** of daily sales volume.

- **Top Day of Week:** Tuesday leads all days with **\$5.09M revenue (26,063 orders)**, followed closely by Sunday (\$4.93M).

5. Regional & Geographic Distribution

| City          | State | Total Sales (\$) | Total Orders | Revenue Share (%) |
|---------------|-------|------------------|--------------|-------------------|
| San Francisco | CA    | \$8,255,000      | 42,898       | 23.95%            |
| Los Angeles   | CA    | \$5,448,000      | 28,498       | 15.81%            |
| New York City | NY    | \$4,662,000      | 23,848       | 13.53%            |
| Boston        | MA    | \$3,659,000      | 19,092       | 10.62%            |
| Atlanta       | GA    | \$2,794,000      | 14,253       | 8.11%             |
| Dallas        | TX    | \$2,765,000      | 14,240       | 8.02%             |
| Seattle       | WA    | \$2,745,000      | 14,119       | 7.96%             |
| Portland      | OR    | \$2,319,000      | 11,980       | 6.73%             |
| Austin        | TX    | \$1,818,000      | 9,509        | 5.27%             |

6. Strategic Business Recommendations

RECOMMENDED ACTION PLAN FOR E-COMMERCE GROWTH

1. Automated Product Bundling & Cross-Selling:

- Pair high-margin hardware with high-frequency accessories at checkout (e.g., offer 20% off Charging Cables or AirPods with every Macbook Pro or iPhone purchase).

2. Precision Hourly Marketing (Dayparting):

- Concentrate search & social media advertising budgets during peak shopping windows: **11:00 AM – 1:00 PM** and **6:00 PM – 8:00 PM** to maximize conversion rates.

3. Regional Supply Chain Optimization:

- Establish primary fulfillment inventory hubs in **San Francisco, Los Angeles, and New York** to enable 1-day shipping for high-value laptops and monitors.

4. Q4 Seasonality Preparation:

- Scale inventory and promotional campaigns starting in October to fully capitalize on the Q4 surge (\$11.54M revenue window).

## 7. Python Analysis Pipeline (Technical Reference)

```
# Core Data Cleaning & Aggregation Pipeline in Python
import pandas as pd

df = pd.read_csv('Amazon_Sales_Data.csv').dropna(how='all')
df['Order Date'] = pd.to_datetime(df['Order Date'], errors='coerce')
df['Month'] = df['Order Date'].dt.month_name()
df['Hour'] = df['Order Date'].dt.hour
df['Quantity Ordered'] = pd.to_numeric(df['Quantity Ordered'], errors='coerce')
df['Price Each'] = pd.to_numeric(df['Price Each'], errors='coerce')
df['Sales'] = df['Quantity Ordered'] * df['Price Each']

df['City'] = df['Purchase Address'].apply(lambda x: x.split(',')[1].strip() if pd.notnull(x) else '')

# Key Aggregations
product_metrics = df.groupby('Product').agg({'Sales': 'sum', 'Quantity Ordered': 'sum'}).sort_values(by='Sales', ascending=False)
city_metrics = df.groupby('City')['Sales'].sum().sort_values(ascending=False)
monthly_trend = df.groupby('Month')['Sales'].sum()
```